

Cyber Security

BRANCH: INFORMATION TECH (IT)
COURSE CODE: IT401

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Cyber Security (IT401) under the Information Tech (IT) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Cryptography Principles

Symmetric Encryption (DES, AES) vs Asymmetric Encryption (RSA, ECC). Hashing algorithms: MD5, SHA-256. Digital Signatures, PKI.

Unit 2: Network & Application Security

Firewalls, Intrusion Detection and Prevention Systems (IDS/IPS). Secure Socket Layer (SSL/TLS), HTTPS. SQL Injection, XSS, CSRF attacks.

Unit 3: Threat Landscape & Hacking

Phishing, Social Engineering, Ransomware, DDoS attacks. Pentesting phases: Reconnaissance, scanning, exploitation, post-exploitation.

Unit 4: Cyber Law & Incident Response

IT Act 2000, GDPR, HIPAA regulations. Digital forensics, chain of custody, log analysis, disaster recovery.

*RSA Key Generation: $n = p * q$, $\Phi(n) = (p-1)*(q-1)$. Encryption: $c = m^e \bmod n$.*

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



